Vidyavardhini's College of Engineering & Technology

Department of Computer Engineering

Academic Year : 2024-25

Class:	**BE**	**Semester:**	**VIII**
Course Code:	**CSL801**	**Course Name:**	**Distributed Computing Lab**

Name of Student:	**Mrudul Parag Chaudhari**
Roll No. :	**10**
Division:	**-**
Experiment No.:	**02**
Title of Experiment:	**To client server using RPC/RMI**
Date of Submission:	
Date of Correction:	

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Performance	5	
Understanding	5	
Journal work and timely submission	10	
Total	20	

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Performance	4-5	2-3	1
Understanding	4-5	2-3	1
Journal work and timely submission	8-10	5-8	1-4

Checked by

Name of Faculty : **Mrs. Swati Saigaonkar**

Signature :

Date :

Aim: To implement Client/Server using RPC/RMI

Objective: Develop a program to implement Client/Server using RPC/RMI

Theory:

The **RMI** (Remote Method Invocation) is an API that provides a mechanism to create distributed application in java. It allows an object to invoke methods on an object running in another JVM. It provides remote communication between the applications using two objects *stub* and *skeleton*.

Understanding stub and skeleton:

RMI uses stub and skeleton object for communication with the remote object. A **remote object** is an object whose method can be invoked from another JVM. Let's understand the stub and skeleton objects:

Stub

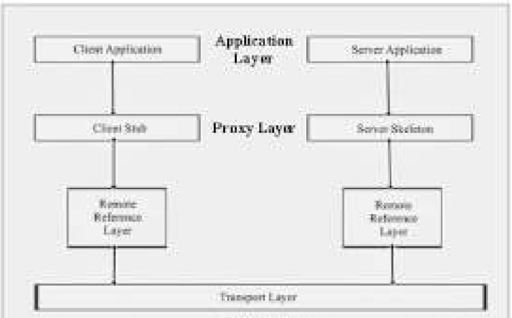
The stub is an object, acts as a gateway for the client side. All the outgoing requests are routed through it. It resides at the client side and represents the remote object. When the caller invokes method on the stub object, it does the following tasks:

1. It initiates a connection with remote Virtual Machine (JVM),
2. It writes and transmits (marshals) the parameters to the remote Virtual Machine (JVM),
3. It waits for the result
4. It reads (unmarshals) the return value or exception, and
5. It finally, returns the value to the caller.

Skeleton

The skeleton is an object, acts as a gateway for the server side object. All the incoming requests are routed through it. When the skeleton receives the incoming request, it does the following tasks:

1. It reads the parameter for the remote method
2. It invokes the method on the actual remote object, and
3. It writes and transmits (marshals) the result to the caller.



The RMI Layers

Steps to write the RMI program

6 steps to write the RMI program.

1. Create the remote interface
2. Provide the implementation of the remote interface
3. Compile the implementation class and create the stub and skeleton objects using the rmic tool
4. Start the registry service by rmiregistry tool
5. Create and start the remote application
6. Create and start the client application

Code and output:

SampleClient.java:-

```
import      java.rmi.*;
import
java.rmi.server.*;
public             class
SampleClient
{ public static void main(String[] args)
    {
      // set the security manager for the client
      System.setSecurityManager(new RMISecurityManager());
      //get the remote object from the registry
      try
        {
         System.out.println("Security Manager loaded");
```

```
      String url = "//localhost/SAMPLE-SERVER";
      SampleServer remoteObject = (SampleServer)Naming.lookup(url);
      System.out.println("Got remote object");
      System.out.println(" 1 + 2 = " + remoteObject.sum(1,2) );
     }


     catch (RemoteException exc) {
      System.out.println("Error in lookup: " + exc.toString()); }
     catch (java.net.MalformedURLException exc) {
      System.out.println("Malformed URL: " + exc.toString());   }
     catch (java.rmi.NotBoundException exc)  {
      System.out.println("NotBound: " + exc.toString());
     }
  }
}
```

SampleServerImpl.java:-

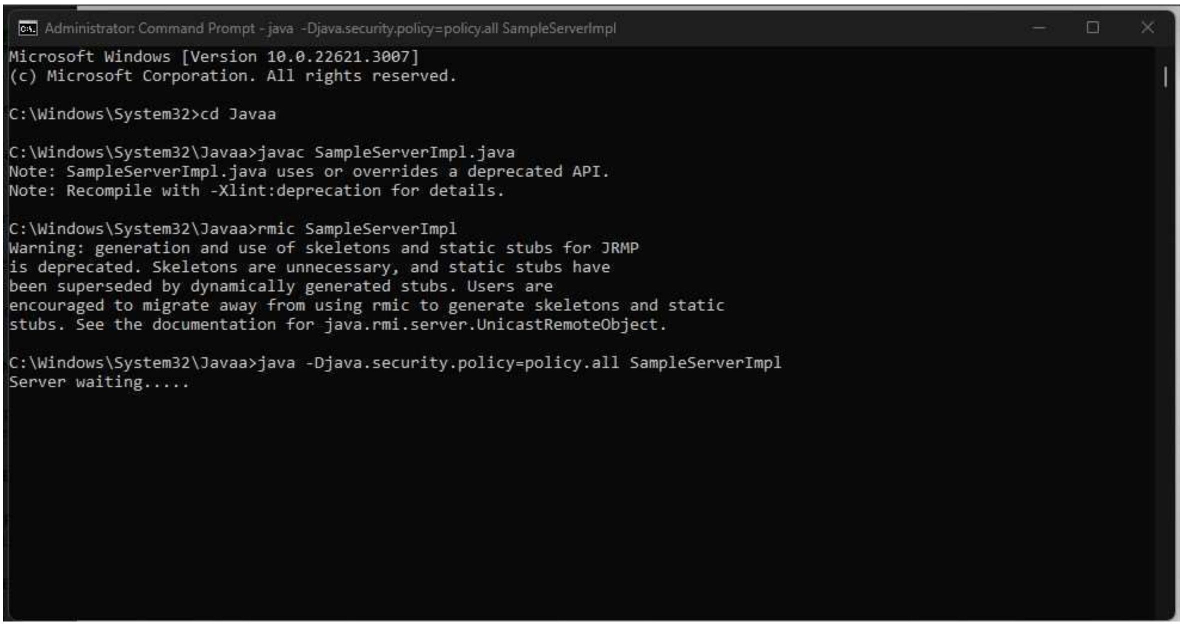
```
import java.rmi.*;
import java.rmi.server.*;
import java.rmi.registry.*;

public class SampleServerImpl extends UnicastRemoteObject implements SampleServer
{
 SampleServerImpl() throws RemoteException
 {


super();
 }

     public int sum(int a,int b) throws RemoteException
     {
            return
     (a+b); }

 public static void main(String args[])
 {
```

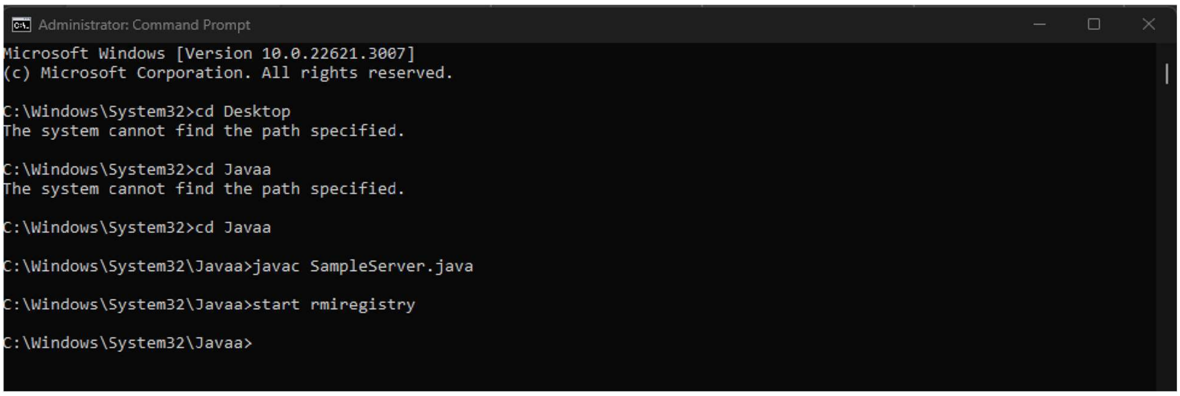
```
    try
    {
     System.setSecurityManager(new    RMISecurityManager());
     //set the security manager

     //create a local instance of the object
     SampleServerImpl Server = new SampleServerImpl();

     //put the local instance in the registry
     Naming.rebind("SAMPLE-SERVER"   ,   Server);   System.out.println("Server
     waiting.....");
    }
    catch (java.net.MalformedURLException me)      {
     System.out.println("Malformed URL: " + me.toString());    }
    catch (RemoteException re) {
     System.out.println("Remote exception: " + re.toString()); }
  }

}
```

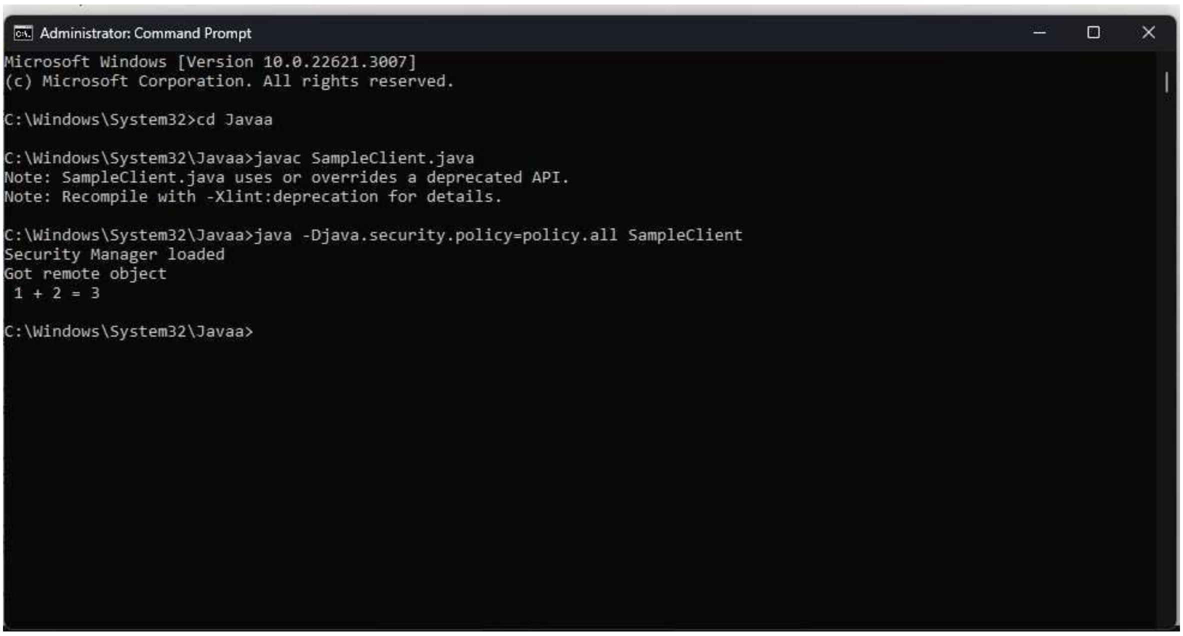
SampleServer.java

```
import java.rmi.*;

public interface SampleServer extends Remote
{   public   int   sum(int   a,int   b)   throws
 RemoteException;
}
```

policy.all:grant { permission
java.security.AllPermission;
 };


Administrator: Command Prompt - java -Djava.security.policy=policy.all SampleServerImpl
Microsoft Windows [Version 10.0.22621.3007]
(c) Microsoft Corporation. All rights reserved.
C:\Windows\System32>cd Javaa
C:\Windows\System32\Javaa>javac SampleServerImpl.java
Note: SampleServerImpl.java uses or overrides a deprecated API.
Note: Recompile with -Xlint:deprecation for details.
C:\Windows\System32\Javaa>rmic SampleServerImpl
Warning: generation and use of skeletons and static stubs for JRMP
is deprecated. Skeletons are unnecessary, and static stubs have
been superseded by dynamically generated stubs. Users are
encouraged to migrate away from using rmic to generate skeletons and static
stubs. See the documentation for java.rmi.server.UnicastRemoteObject.
C:\Windows\System32\Javaa>java -Djava.security.policy=policy.all SampleServerImpl
Server waiting.....





Conclusion: Implementing Client/Server architecture using Remote Procedure Call (RPC) or Remote Method Invocation (RMI) offers a robust framework for distributed computing. RPC/RMI facilitates seamless communication between client and server, abstracting the complexities of network programming. It promotes modularity, scalability, and maintainability by allowing developers to encapsulate functionality into remote methods. While RPC/RMI introduces overhead due to network communication, its advantages in code organization and ease of development outweigh these concerns in many scenarios. Overall, RPC/RMI serves as a foundational technology for building efficient and reliable distributed systems.